

OFFICIAL ARCHITECTURE & ENGINEERING DOCUMENTATION

Smart School Management System

Complete System Architecture, Flowcharts, Database Entity-Relationship Specification & Module Index

PROJECT VERSION	RELEASE DATE
v1.5 (Production Ready)	September 29, 2026
TECHNOLOGY STACK	MODULES & TABLES
Vanilla JS SPA + CodeIgniter 3.x + MySQL 8.x	45 Modules / 56 Relational Tables
DEVELOPED BY	TARGET DEPLOYMENT
Infosof Technologies	cPanel Shared Hosting / Linux VPS

Smart School Management System

Comprehensive Project Report

Infosof Technologies | Academic Year 2026–27

[!IMPORTANT] **Project:** Smart School Management System
Version: 1.5 (September 2026)
Stack: HTML/CSS/Vanilla JS (SPA) + CodeIgniter 3.x (REST API) + MySQL 8.x
Developer: Infosof Technologies
Institution: Smart School International, Kolkata

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1. Executive Summary

Smart School is a **full-stack, web-based school management system** that digitises and centralises all administrative, academic, financial, and operational workflows of a school. It supports **six user roles** (Super Admin, Admin, Teacher, Accountant, Parent, Student), provides a **REST JSON API backend** powered by CodeIgniter 3, and a **SPA (Single Page Application) frontend** written in plain HTML/CSS/JavaScript.

Key Highlights

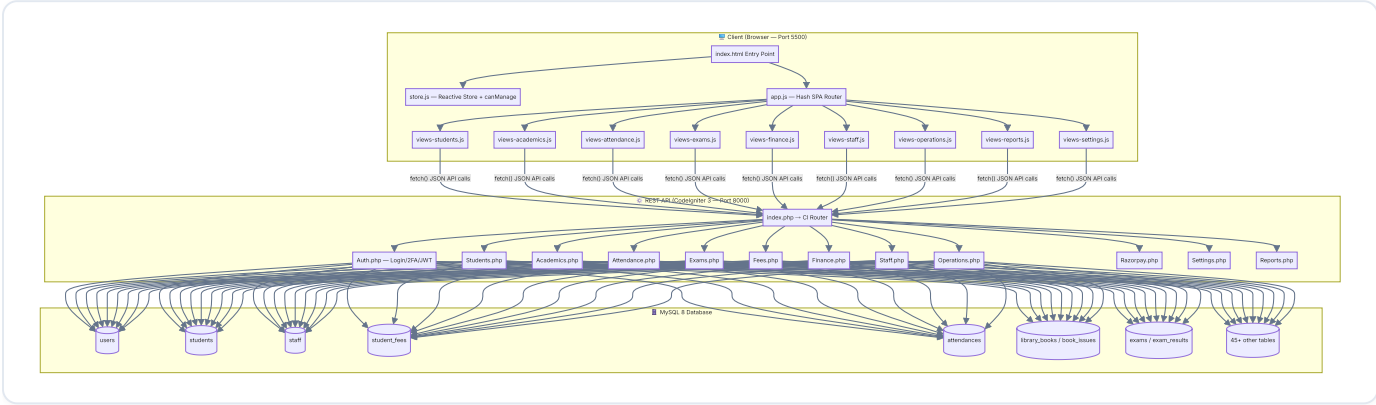
Metric	Value
Total Modules	45
Database Tables	56
User Roles	6 (Super Admin, Admin, Teacher, Accountant, Librarian, Receptionist, Parent, Student)
API Endpoints	45+
Payment Gateway	Razorpay (Online + Offline)

METRIC	VALUE
Attendance Modes	Manual + QR Code Scan
Deployment Target	cPanel Shared Hosting / VPS

2. Technology Stack

TECHNOLOGY STACK	
Layer	Technology
Frontend UI	HTML5, Vanilla CSS3, Vanilla JavaScript (ES2020)
SPA Router	Hash-based routing (#/path) in app.js
State Store	window.SS_STORE (localStorage reactive engine)
RBAC	window.canManage() – token-based role check
Backend API	PHP 8.3 + CodeIgniter 3.x (REST JSON API)
Database	MySQL 8.x – 56 tables, InnoDB, utf8mb4
Payment	Razorpay SDK (test + live mode)
Web Server	Apache 2.4 with mod_rewrite (.htaccess)
Dev Server	Node.js static file server (port 5500)
PHP Server	PHP built-in server (port 8000 for API)

3. System Architecture



File Structure

```

Smart-School-Management-System/
├── frontend/
│   ├── index.html
│   ├── css/
│   │   └── style.css
│   ├── js/
│   │   ├── store.js
│   │   ├── app.js
│   │   ├── views-students.js
│   │   ├── views-academics.js
│   │   ├── views-attendance.js
│   │   ├── views-exams.js
│   │   ├── views-finance.js
│   │   ├── views-staff.js
│   │   ├── views-operations.js
│   │   ├── views-reports.js
│   │   └── views-settings.js
│   └── assets/
│       └── favicon.jpg
├── backend_codeigniter/
│   ├── application/
│   │   ├── controllers/
│   │   ├── models/
│   │   └── config/
│   │       ├── database.php
│   │       └── routes.php
├── database/
│   └── smart_school_database_cpanel_2026.sql
├── index.php
├── .htaccess
└── .env.example

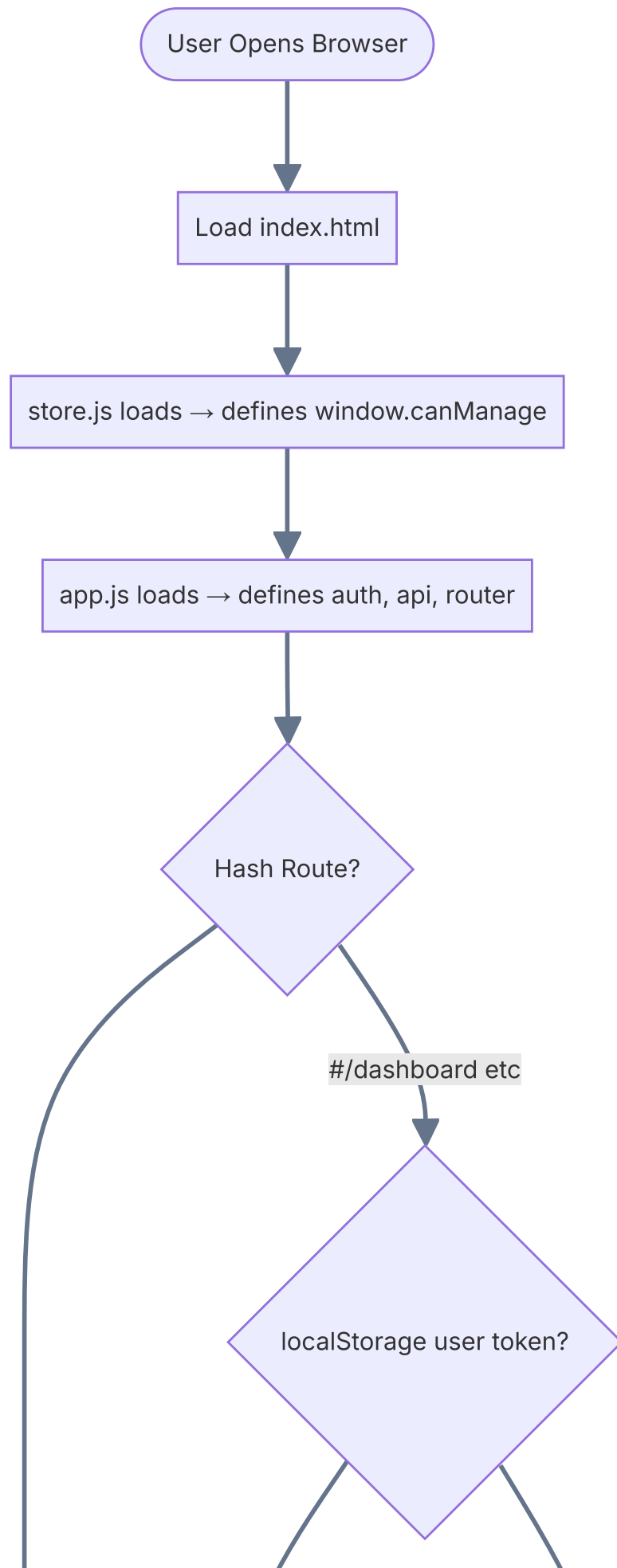
```

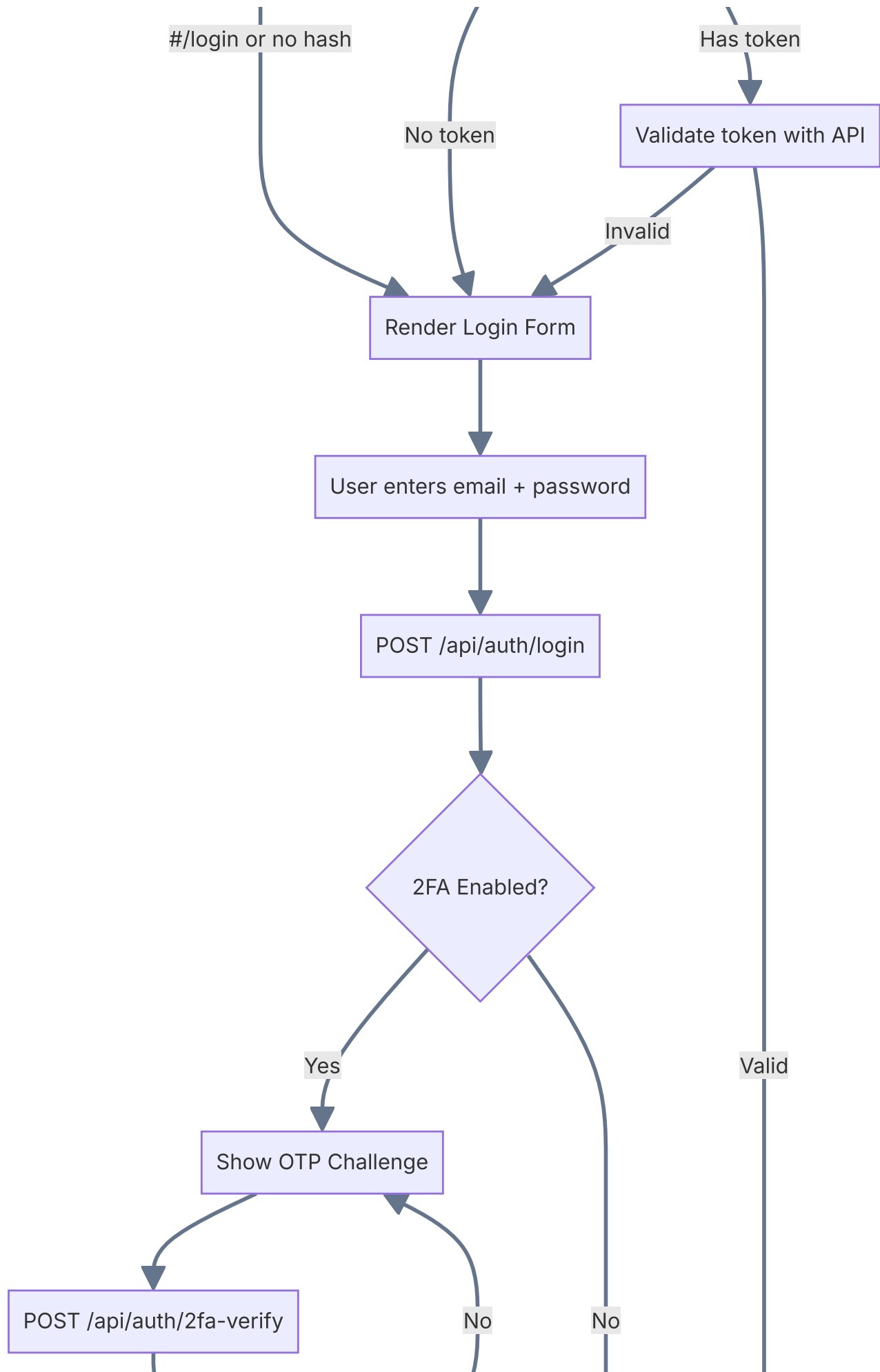
← SPA Frontend
 ← Entry point (loads all scripts)
 ← Complete design system
 ← Reactive store, canManage(), modal engine
 ← SPA router, auth, API client, nav
 ← Students, Admission, TC, Credentials
 ← Calendar, Downloads, Timetable, Live Classes
 ← Mark, QR, Report
 ← Exams, Marks Entry, Admit Cards
 ← Fees, Payroll, Accounts, Razorpay
 ← Staff directory, Leave, Payslips
 ← Library, Transport, Hostel, Notices
 ← Analytics and Reports
 ← School settings, RBAC, Custom Fields
 ← REST API Backend
 ← 15 controllers
 ← 10 models
 ← MySQL connection
 ← API route mappings
 ← Full DB dump (56 tables)
 ← Apache root → CI bootstrap
 ← mod_rewrite rules
 ← Environment config template

4. Application Working Flow

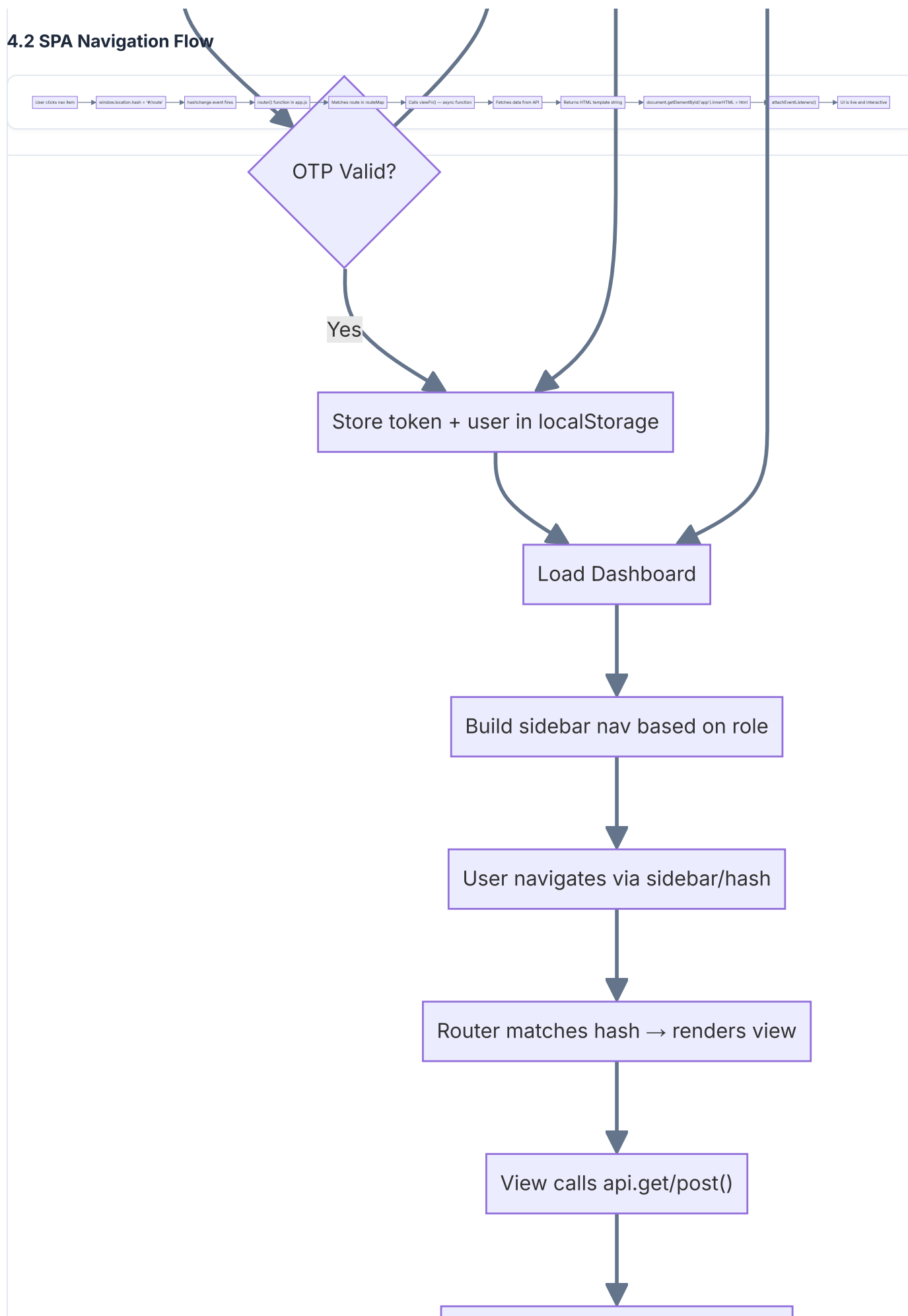
4.1 Authentication Flow







4.2 SPA Navigation Flow



5. Module Flowcharts

5.1 Fee Collection Flow (with Razorpay)

API attaches Bearer token header



CI Backend validates token

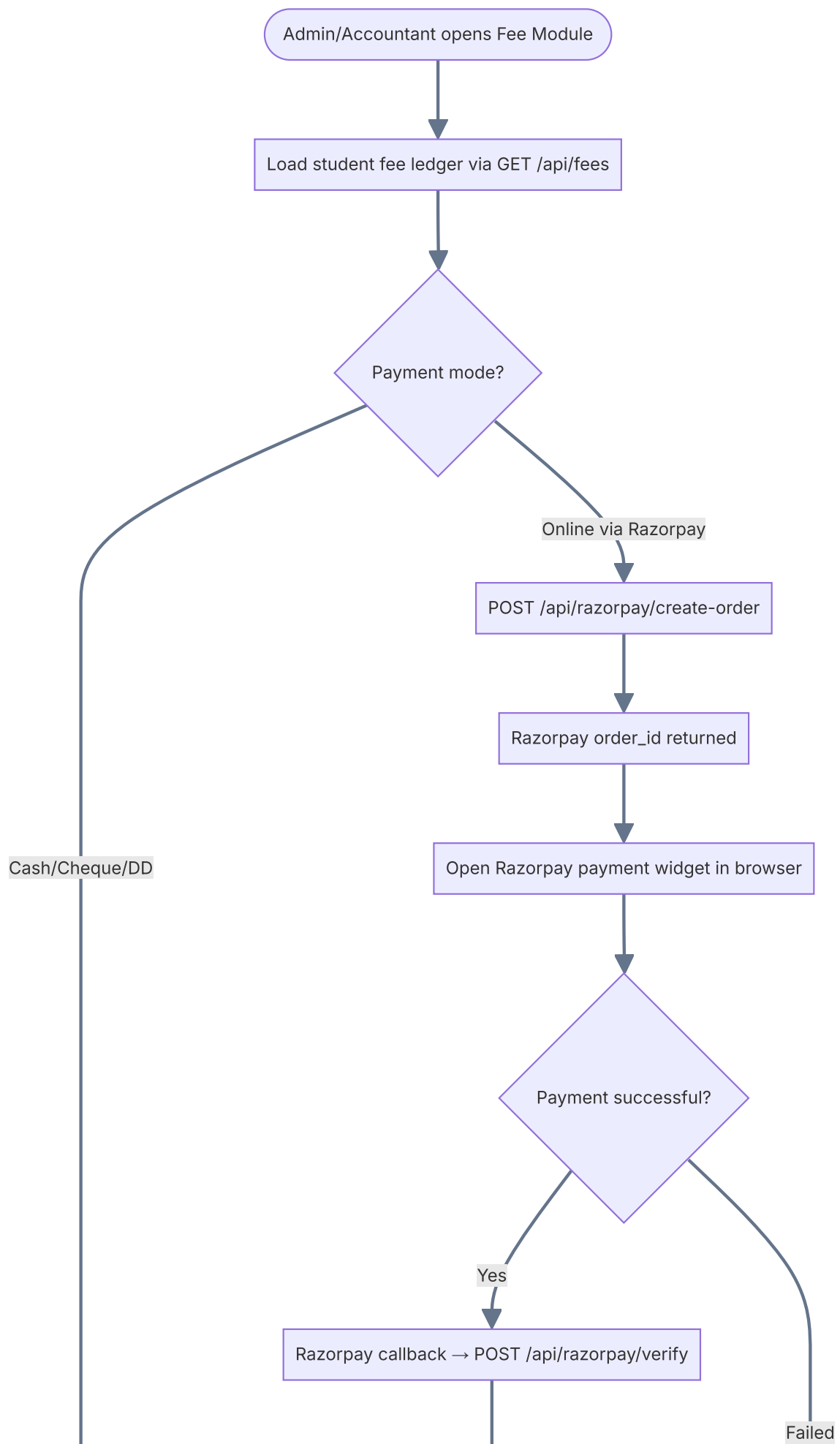


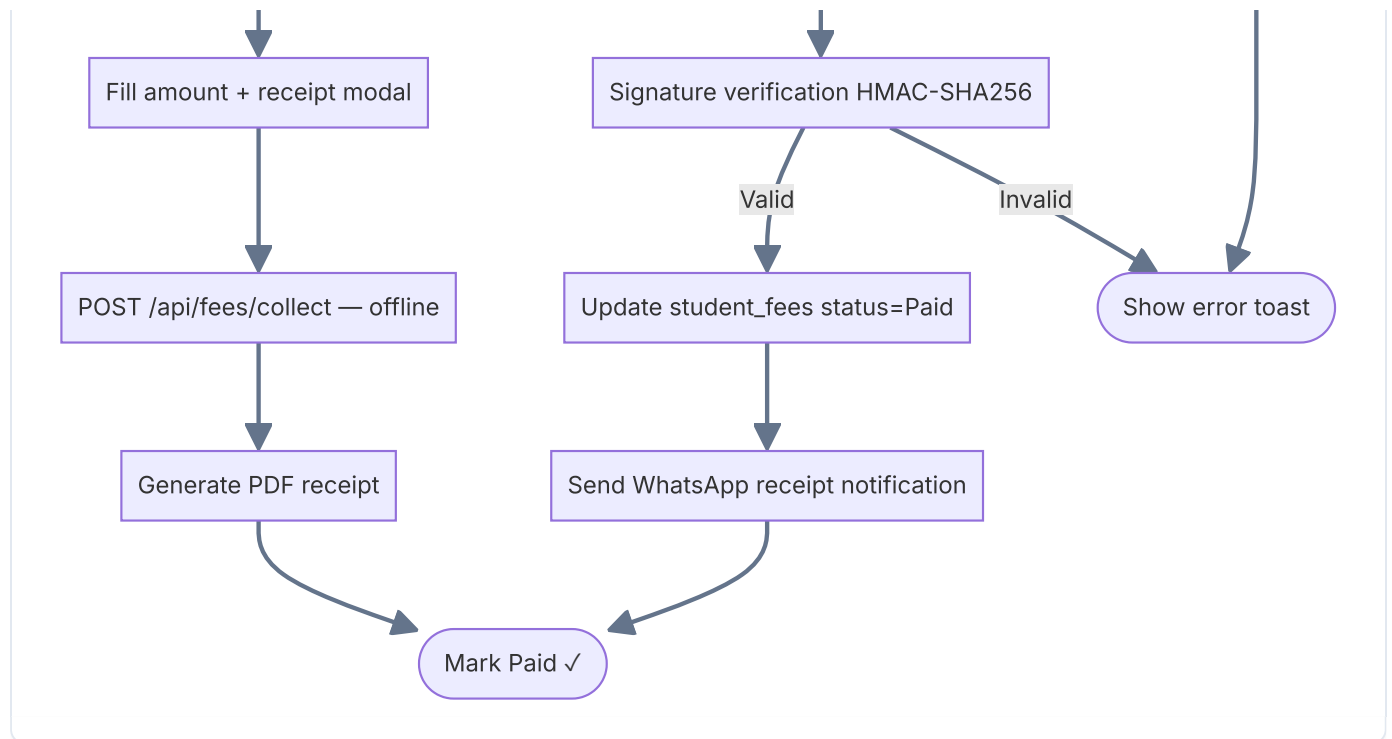
Return JSON data



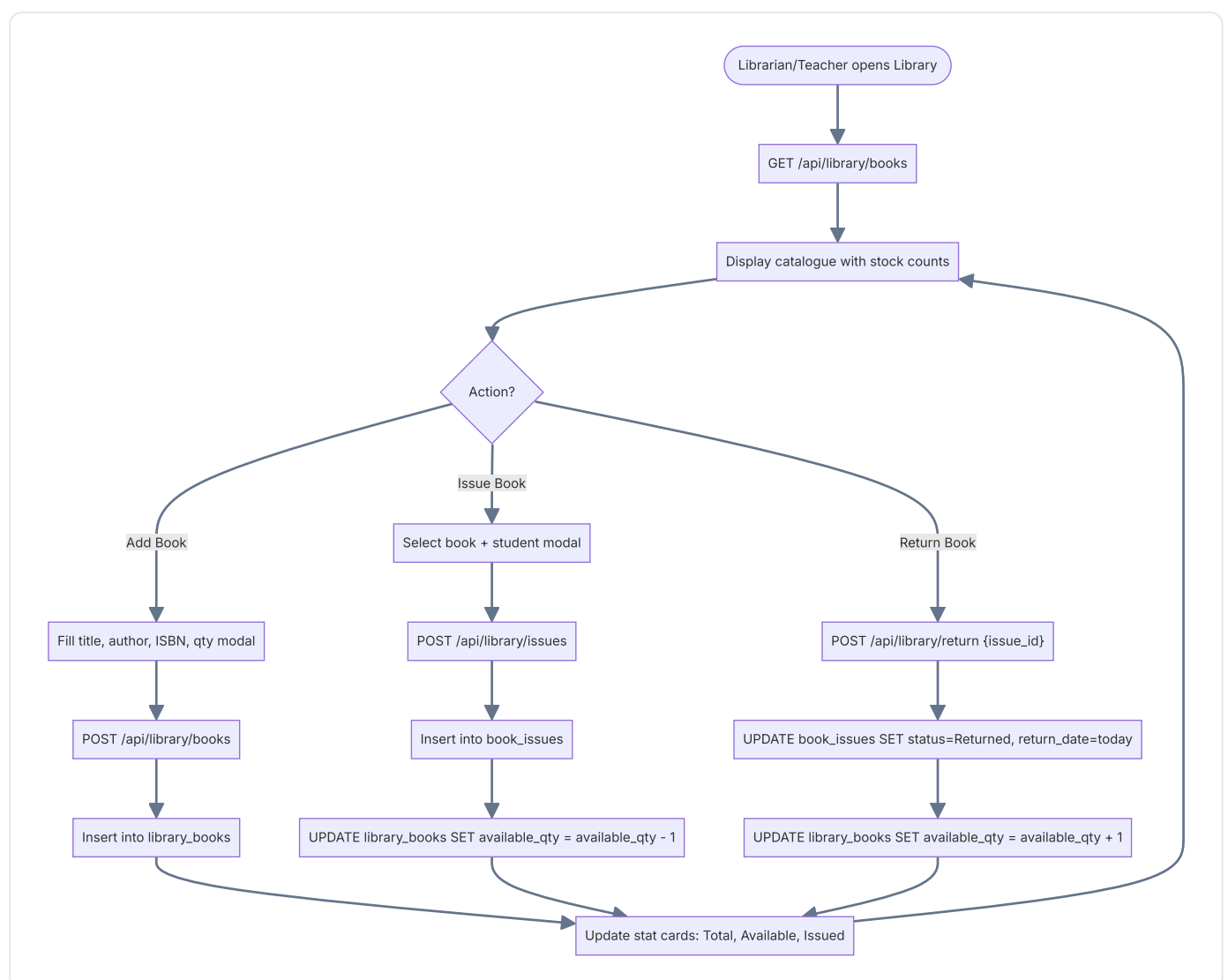
View renders DOM



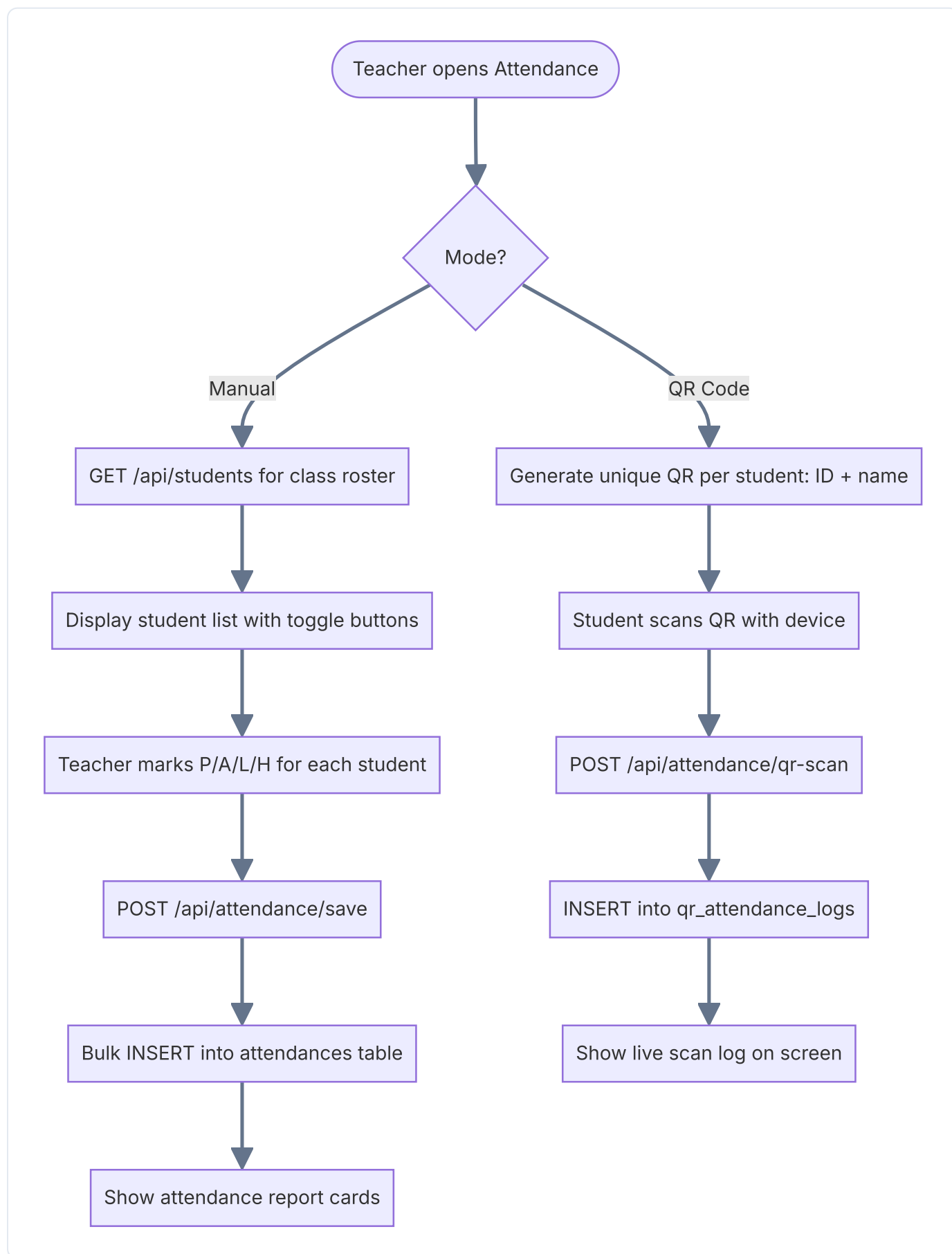




5.2 Library Book Issue & Return Flow

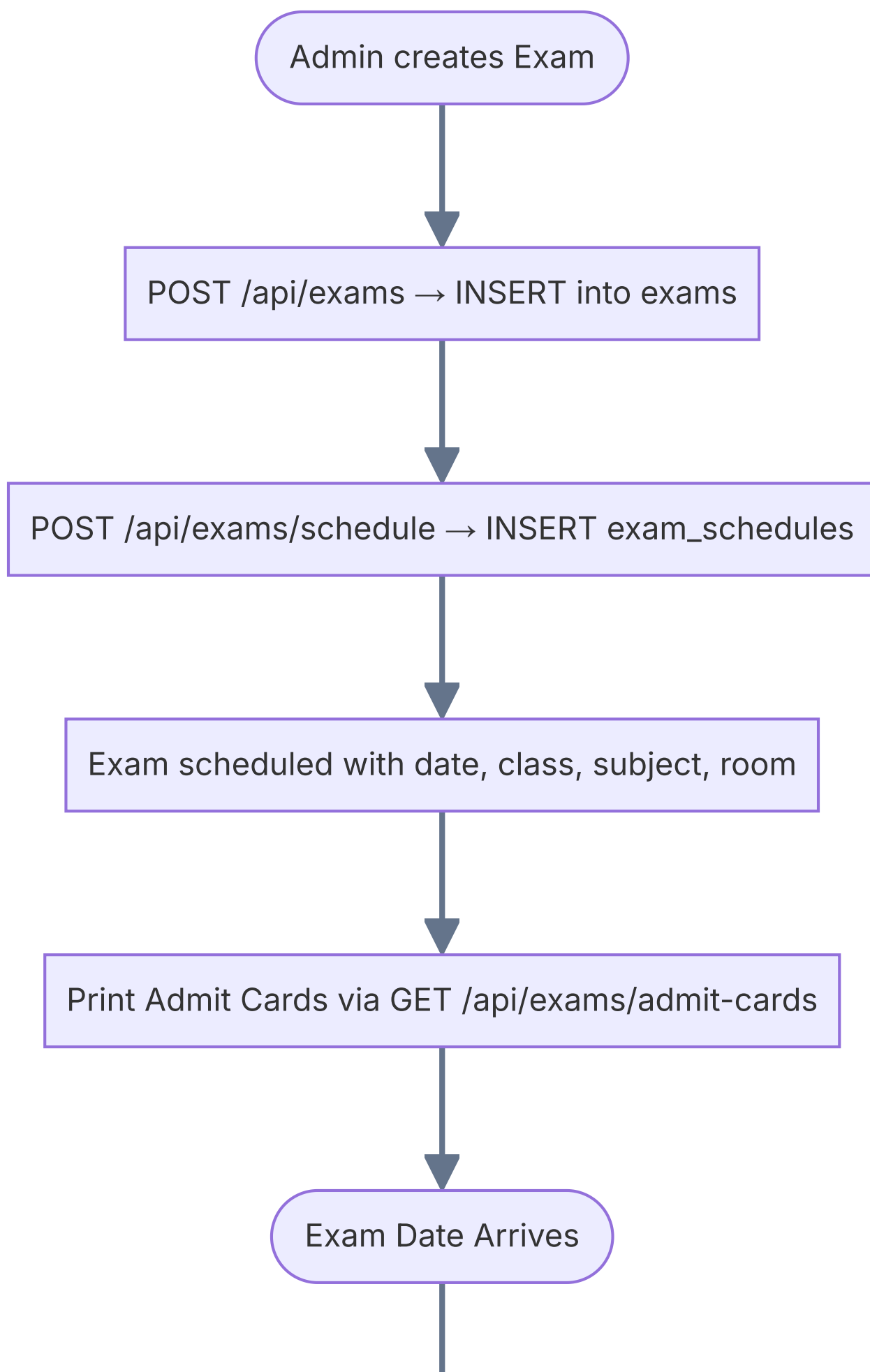


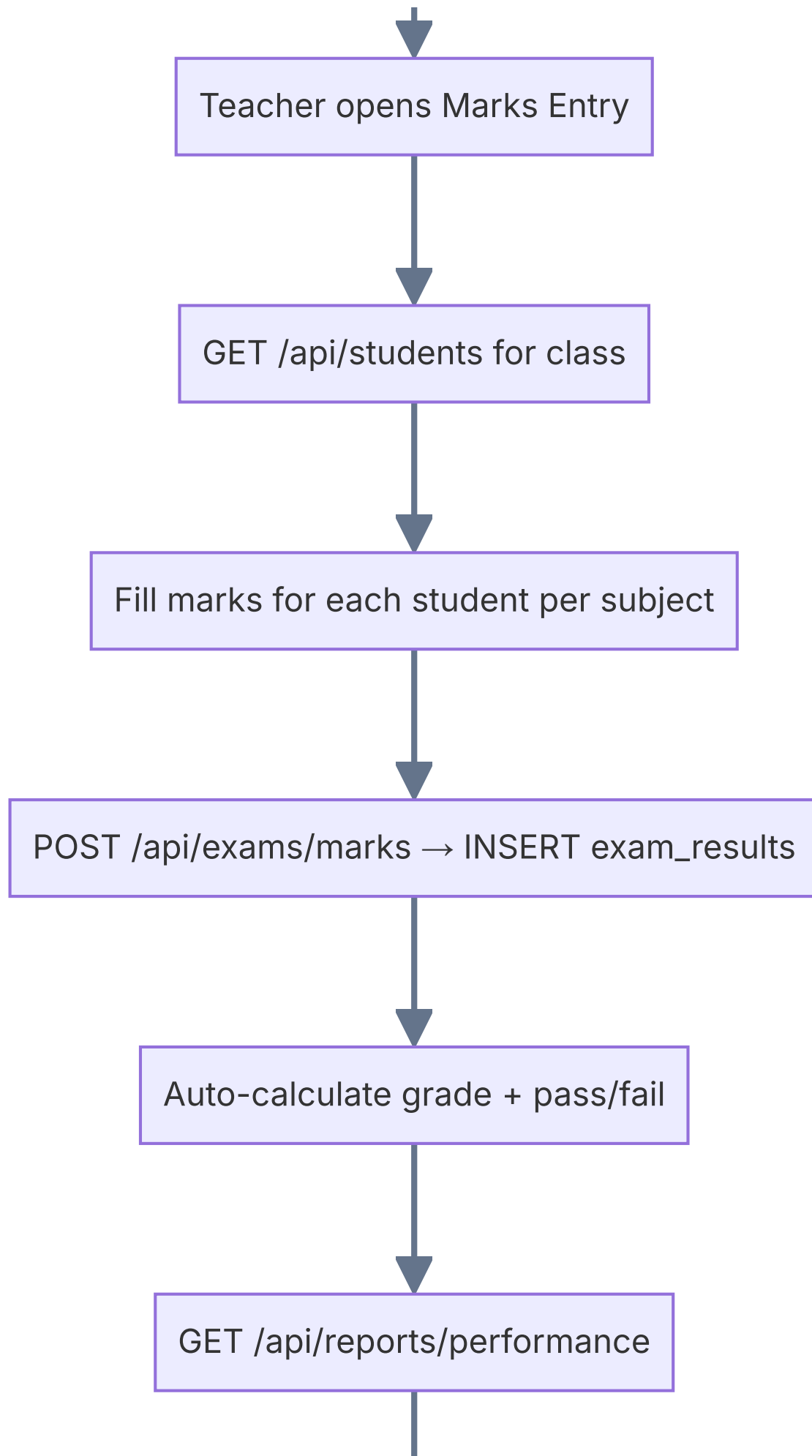
5.3 Attendance Flow (Manual + QR)



5.4 Exam & Marks Flow





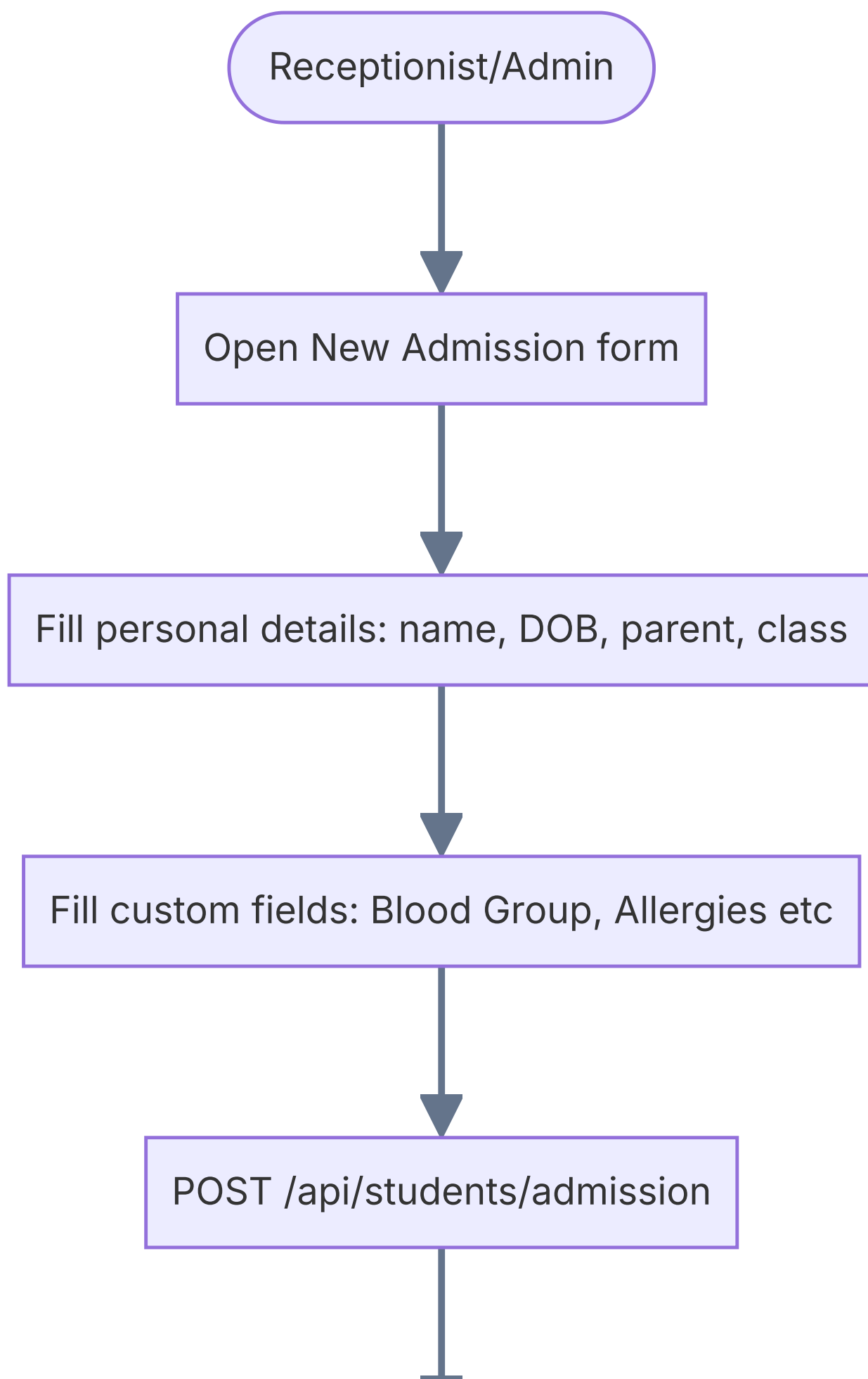


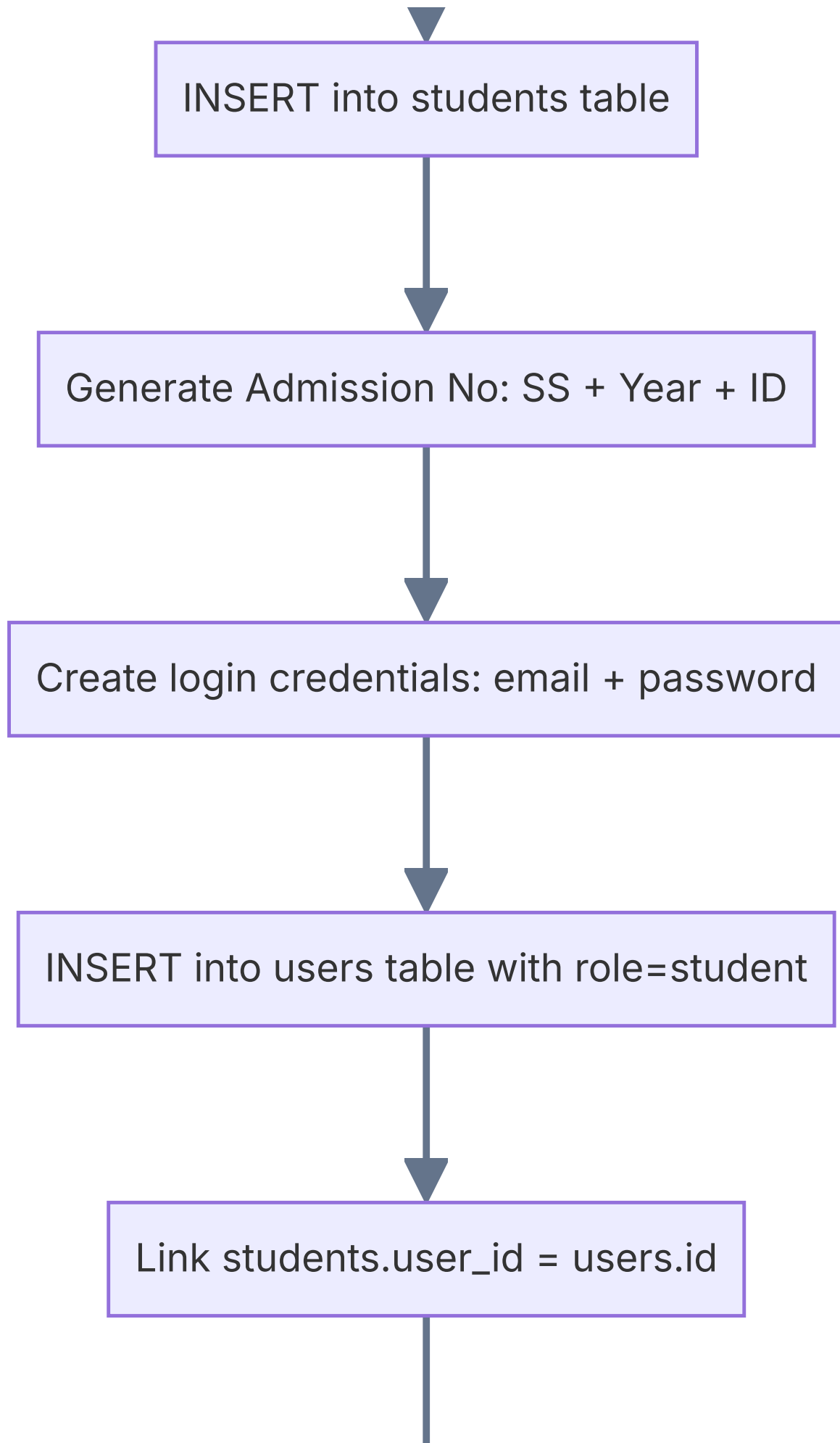
5.5 Student Admission Flow

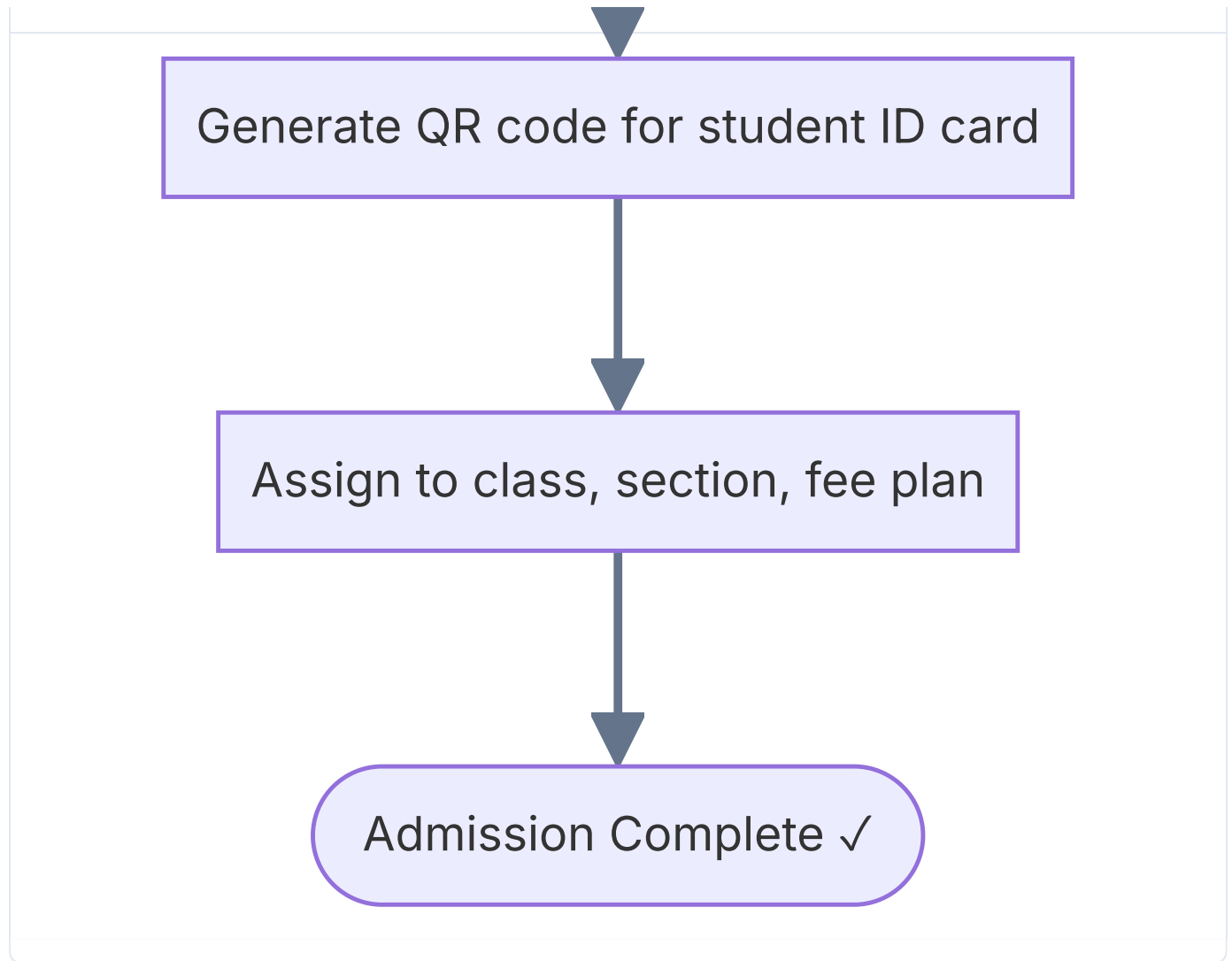


Generate performance analytics charts





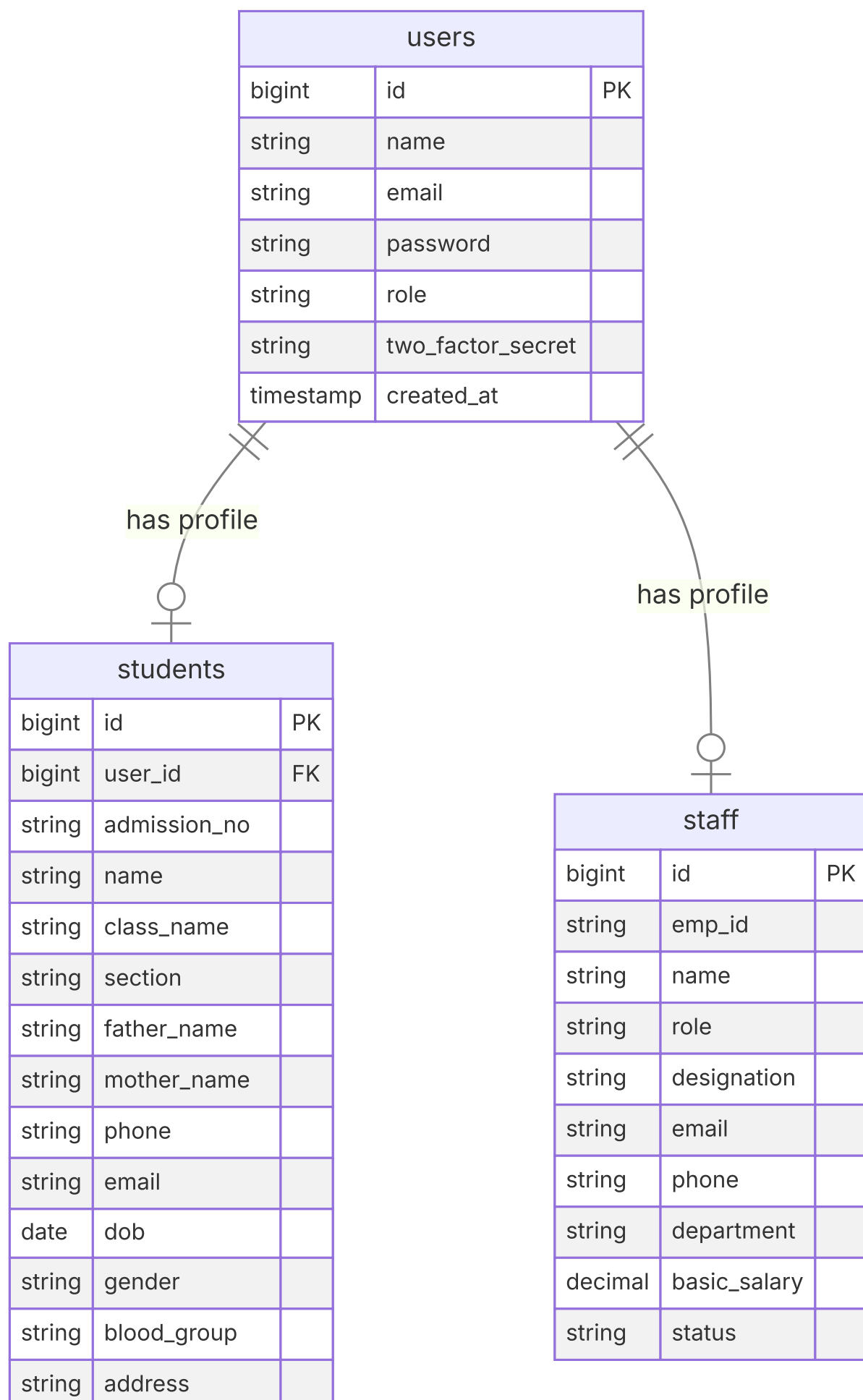


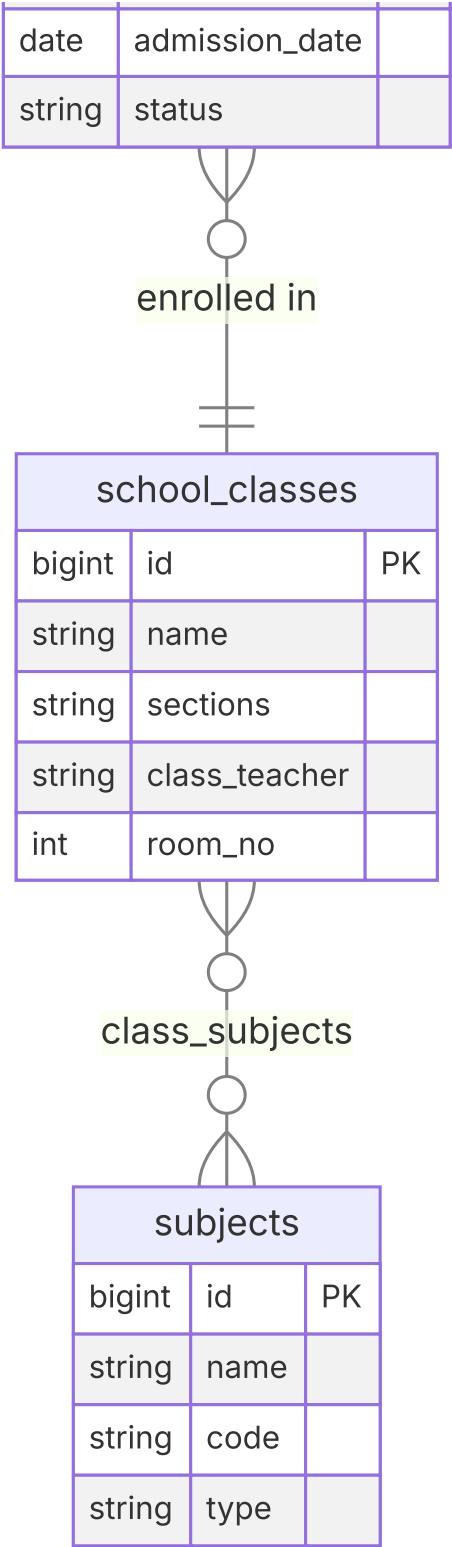


6. Database ER Diagram

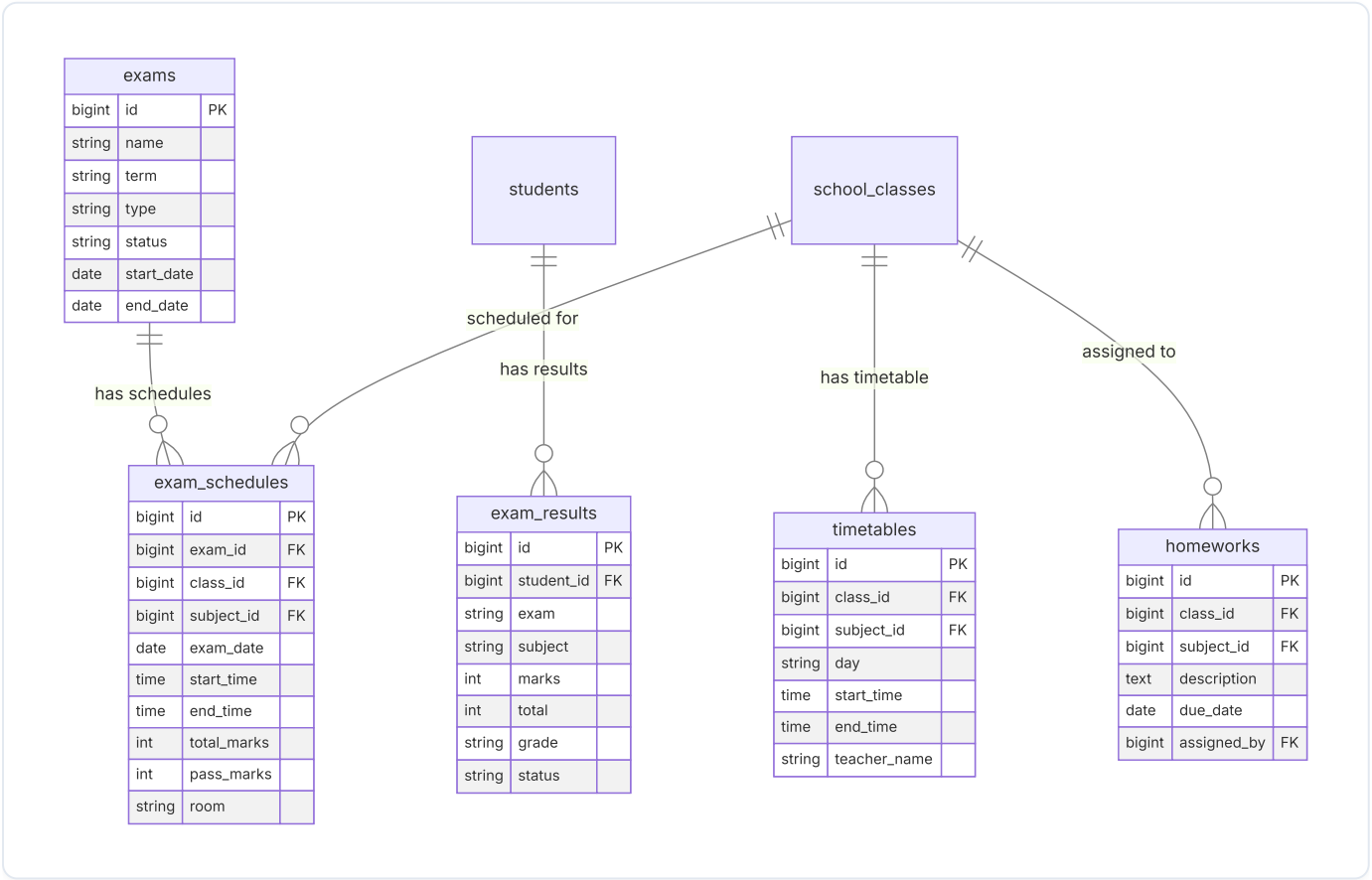
6.1 Core Entity Relationships



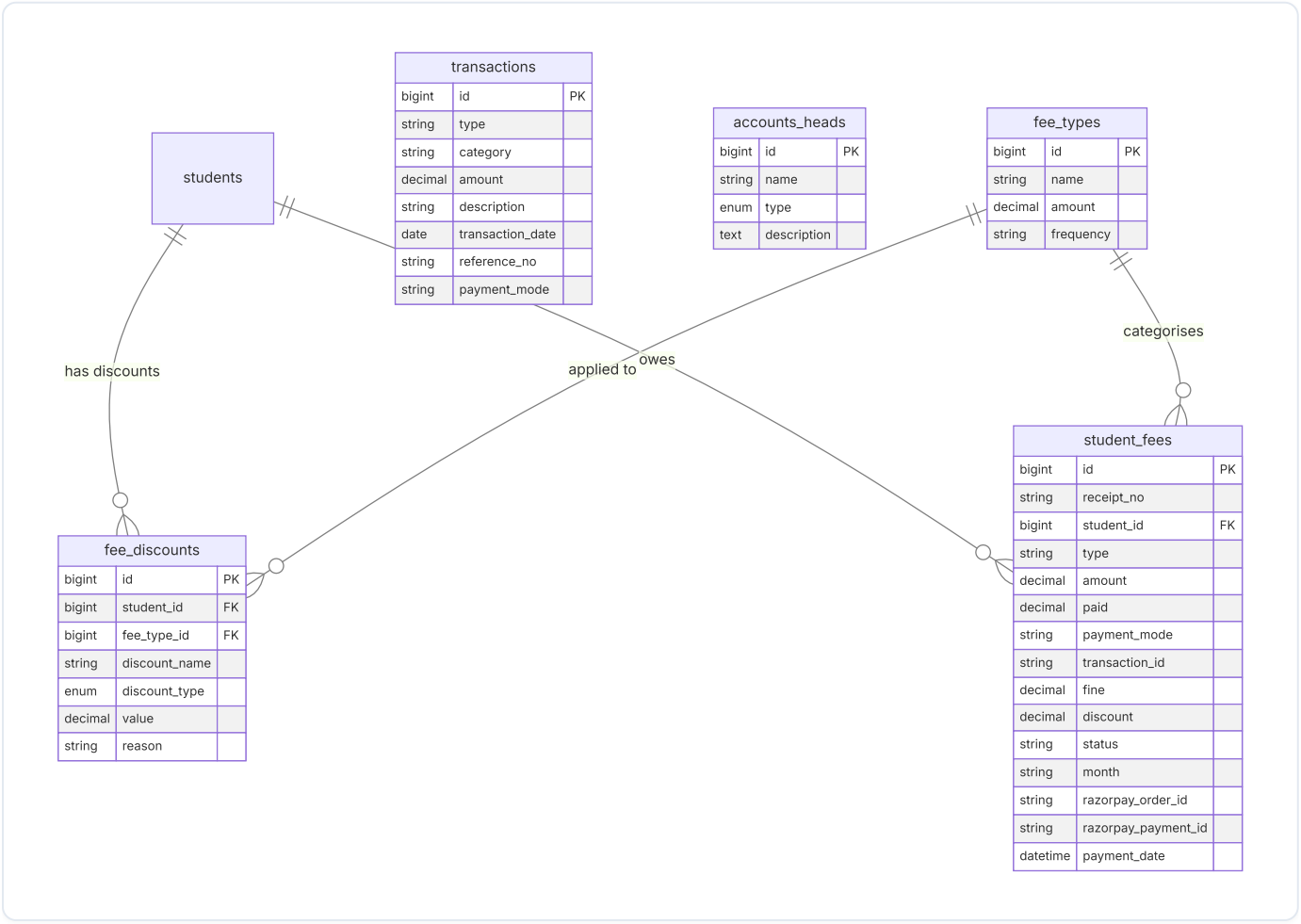




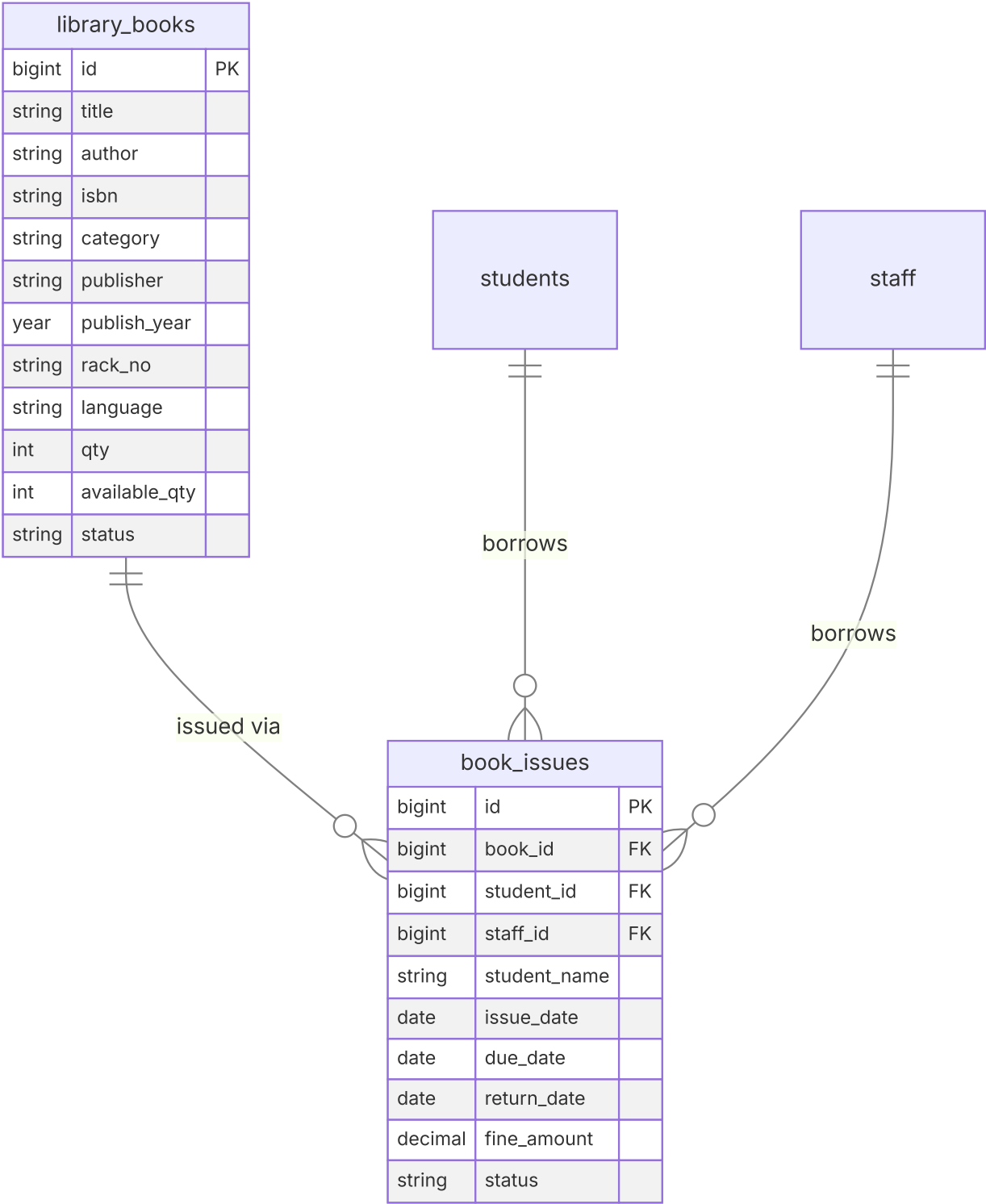
6.2 Academic & Examination Relationships



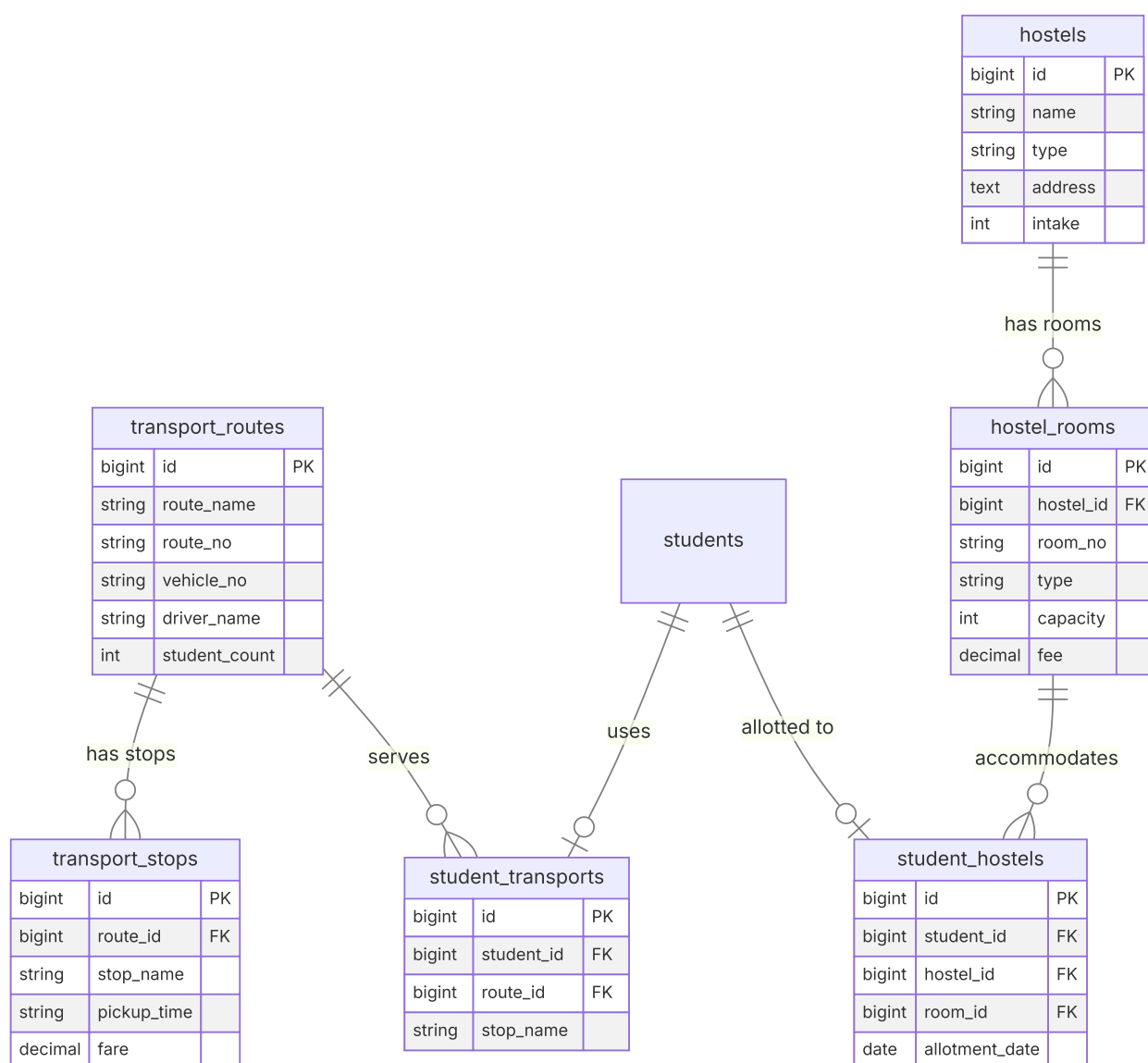
6.3 Finance & Fee Relationships



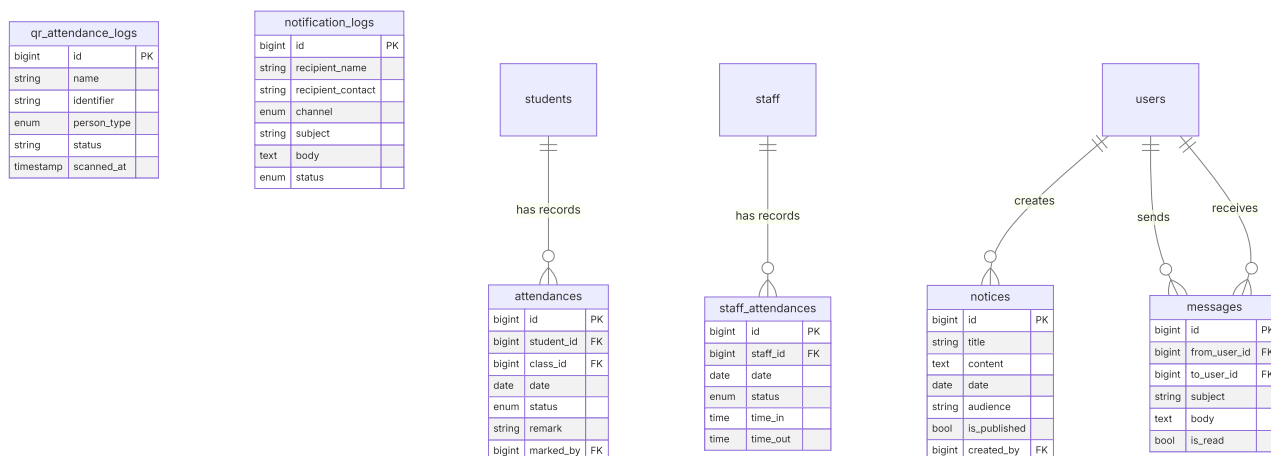
6.4 Library Relationships



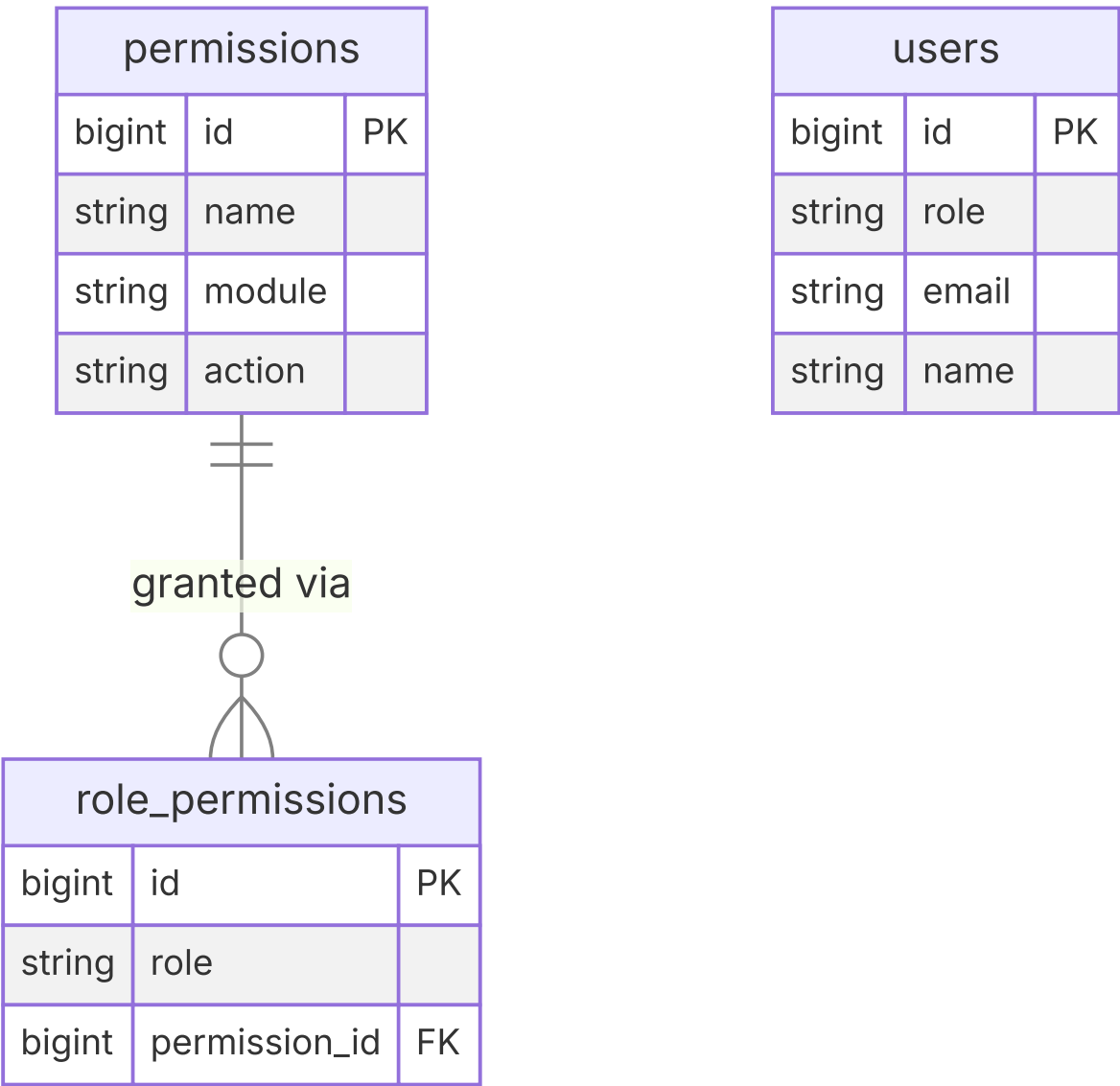
6.5 Operations Relationships



6.6 Attendance & Communication Relationships



6.7 RBAC Relationships



7. Role-Based Access Control (RBAC)

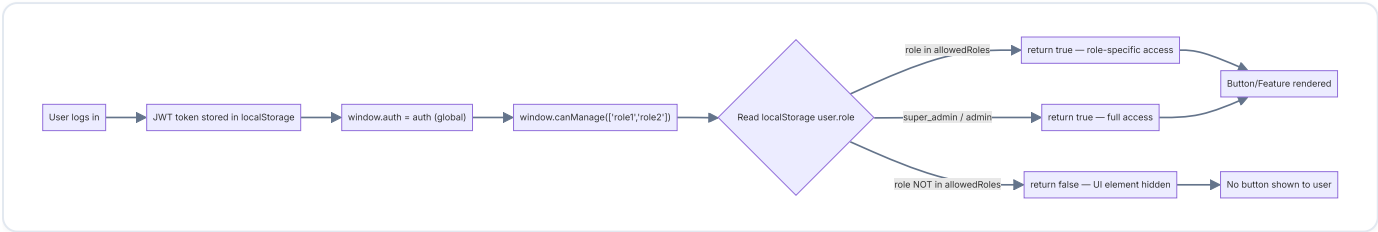
Role Permission Matrix

MODULE / ACTION	SUPER ADMIN	ADMIN	TEACHER	ACCOUNTANT	LIBRARIAN	RECEPTIONIST	PARENT	STUDENT
Students — View	✓	✓	✓			✓		
Students — Create/Edit/Delete	✓	✓				✓		
New Admission	✓	✓				✓		

MODULE / ACTION	SUPER ADMIN	ADMIN	TEACHER	ACCOUNTANT	LIBRARIAN	RECEPTIONIST	PARENT	STUDENT
Transfer Certificate	✓	✓						
Staff — View	✓	✓						
Staff — Create/Edit/Delete	✓	✓						
Mark Attendance	✓	✓	✓					
QR Attendance	✓	✓	✓			✓		
Attendance Report	✓	✓	✓					
Examinations	✓	✓	✓				✓	✓
Marks Entry	✓	✓	✓					
Admit Card	✓	✓	✓				✓	✓
Fee Collect / View	✓	✓		✓				
Razorpay Payments	✓	✓		✓				
Annual Calendar — View	✓	✓	✓				✓	✓
Annual Calendar — Add/Delete	✓	✓	✓				✗	✗
Download Center — View/Download	✓	✓	✓				✓	✓
Download Center — Upload/Delete	✓	✓	✓				✗	✗
Live Classes — View/Join	✓	✓	✓				✓	✓
Live Classes — Schedule/Delete	✓	✓	✓				✗	✗
Library — View	✓	✓	✓		✓			✓
Library — Issue/Return/Add	✓	✓	✓		✓			✗
Transport — View	✓	✓					✓	✓
Transport — Manage	✓	✓					✗	✗
Hostel — View/Manage	✓	✓						
Notices — View	✓	✓	✓				✓	✓

MODULE / ACTION	SUPER ADMIN	ADMIN	TEACHER	ACCOUNTANT	LIBRARIAN	RECEPTIONIST	PARENT	STUDENT
Notices — Create/Edit	✓	✓	✓					
Payroll — View	✓	✓		✓				
Payroll — Generate/Pay	✓	✓		✓				
Reports	✓	✓		✓				
Settings	✓							

RBAC Implementation



8. API Architecture

Base URL & Structure

```
Backend: http://127.0.0.1:8000/api/
Frontend: http://127.0.0.1:5500/frontend/
```

Key API Endpoints

METHOD	ENDPOINT	CONTROLLER	DESCRIPTION
POST	/api/auth/login	Auth	Login with email + password
POST	/api/auth/2fa-verify	Auth	Two-factor OTP verification
POST	/api/auth/logout	Auth	Invalidate token
GET	/api/students	Students	List all students
POST	/api/students	Students	Create new student
PUT	/api/students/{id}	Students	Update student record
DELETE	/api/students/{id}	Students	Delete student
GET	/api/staff	Staff	List all staff members
POST	/api/attendance/save	Attendance	Bulk save attendance
POST	/api/attendance/qr-scan	Attendance	Log QR scan
GET	/api/attendance/report	Attendance	Attendance analytics

METHOD	ENDPOINT	CONTROLLER	DESCRIPTION
GET	/api/exams	Exams	List examinations
POST	/api/exams/marks	Exams	Save marks batch
GET	/api/fees	Fees	Student fee ledger
POST	/api/fees/collect	Fees	Record offline payment
POST	/api/razorpay/create-order	Razorpay	Create online payment
POST	/api/razorpay/verify	Razorpay	Verify payment signature
GET	/api/library/books	Operations	Book catalogue
POST	/api/library/books	Operations	Add new book
POST	/api/library/issues	Operations	Issue book to student
POST	/api/library/return	Operations	Return book
GET	/api/library/issues	Operations	Circulation ledger
GET	/api/transport/routes	Operations	Bus routes
GET	/api/hostel/rooms	Operations	Hostel rooms
GET	/api/calendar-events	Academics	Annual calendar
POST	/api/calendar-events	Academics	Add event
GET	/api/download-materials	Academics	Study materials
POST	/api/download-materials	Academics	Upload material
GET	/api/live-classes	Academics	Live class schedule
GET	/api/notices	Operations	Notice board
GET	/api/reports/dashboard	Dashboard	KPI dashboard stats
GET	/api/settings	Settings	School configuration
POST	/api/settings	Settings	Update settings
GET	/api/payroll	Finance	Staff payroll
POST	/api/payroll/generate	Finance	Generate salary

API Response Format

```
{
  "status": "success",
  "data": [...],
  "message": "Operation completed successfully",
  "total": 42
}
```

9. Module Catalogue (45 Modules)

Student Management

1. **Student Directory** — Full profile, search, filters
2. **New Admission** — Complete admission workflow with custom fields
3. **Login Credentials** — Generate username/password for students/parents
4. **Student Promotion** — Bulk class promotion at year end
5. **Transfer Certificate** — Automated TC generation
6. **Behavior Records** — Incident logging per student
7. **Student Documents** — Upload & manage certificates, birth cert, etc.

Staff Management

8. **Staff Directory** — Complete HR records
9. **Staff Attendance** — Daily time-in/time-out with status
10. **Leave Applications** — Leave approval workflow
11. **Payroll** — Salary generation, payslips, payment tracking

Academic Management

12. **Classes & Sections** — Class structure management
13. **Subjects** — Subject catalogue with class mapping
14. **Academic Sessions** — Year management (active session)
15. **Timetable** — Weekly class timetable per class
16. **Annual Calendar** — School events, holidays, exams
17. **Homework** — Assignment distribution per class/subject

Attendance

18. **Mark Attendance** — Manual per-student P/A/L/H
19. **QR Code Attendance** — Scan-based attendance with live log
20. **Attendance Report** — Class-wise and student-wise analytics

Examinations

21. **Examinations** — Create and schedule exams
22. **Marks Entry** — Subject-wise marks, auto-grade
23. **Admit Cards** — Student admit card generation and print
24. **Results** — Result cards with grade analysis

Finance

25. **Fee Types** — Tuition, Transport, Hostel, Library fees
26. **Fee Collection** — Per-student payment with receipt
27. **Fee Discounts** — Sibling, merit, and scholarship discounts
28. **Razorpay Online Payment** — Integrated online payment gateway
29. **Accounts / Transactions** — Income/expense ledger
30. **Financial Reports** — Revenue analytics and export

Library

31. **Book Catalogue** — Full inventory with stock tracking
32. **Book Issue** — Issue with due date and fine calculation
33. **Book Return** — Return and auto-increment available stock

34. Circulation Ledger — All issued/returned history

Operations

- 35. **Transport Routes** — Bus routes with stops and fares
- 36. **Hostel** — Building, rooms, student allotments
- 37. **Circular / Notices** — Published announcements per audience
- 38. **Noticeboard** — School-wide notice management

Digital Learning

- 39. **Download Center** — Study material upload and download
- 40. **Live Classes** — Google Meet / Zoom class scheduling and join

Reports & Analytics

- 41. **Dashboard KPIs** — Real-time summary cards
- 42. **Attendance Analytics** — Charts and export
- 43. **Financial Analytics** — Revenue, collection rates
- 44. **Academic Performance** — Student rank, grade distribution

System Administration

- 45. **School Settings** — Profile, campus, Razorpay config, WhatsApp, RBAC, Custom Fields

10. Key Design Decisions

10.1 SPA Architecture

The frontend is a **Single Page Application** using **hash-based routing** (`#/students` , `#/exams` , etc.) with no framework. This decision was made for:

- Zero build step needed — works as static files
- Easy cPanel deployment (no Node.js server needed for frontend)
- Fast navigation without page reloads

10.2 Reactive Store

`window.SS_STORE` provides localStorage-backed reactive state that:

- Persists data across page refreshes as a fallback to the API
- Allows optimistic UI updates before API confirms
- Reduces redundant API calls with in-memory caching

10.3 Token-Based Auth with 2FA

- JWT-style Bearer token stored in `localStorage`
- All API requests attach `Authorization: Bearer <token>` header
- Optional TOTP-based two-factor authentication for admin accounts

10.4 Offline Fallback Strategy

Each view function follows this pattern:

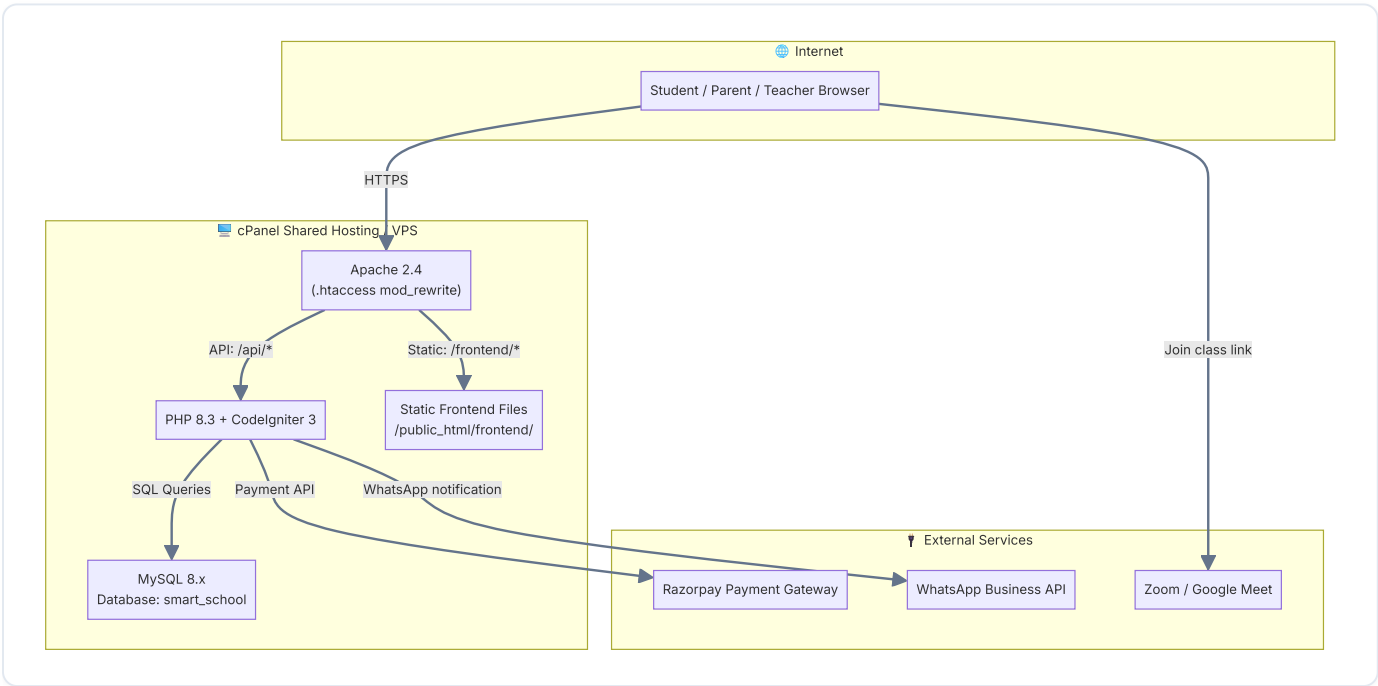
1. Try to fetch from MySQL via REST API
2. On API failure → fall back to `SS_STORE` localStorage data

3. UI always renders — never blank due to API failure

10.5 Razorpay Payment Integration

- Server creates `order_id` via Razorpay API (backend secret key protected)
- Frontend opens Razorpay checkout widget
- On success, `payment_id` + `signature` sent to backend for HMAC-SHA256 verification
- Only after verification is `student_fees.status` updated to `Paid`

11. Deployment Architecture



Environment Configuration (`.env`)

```
DB_HOST=localhost
DB_DATABASE=smart_school
DB_USERNAME=smart_school_user
DB_PASSWORD=<secure_password>

RAZORPAY_KEY_ID=rzp_live_XXXXXXX
RAZORPAY_KEY_SECRET=XXXXXXXXXXXXXXXXX
RAZORPAY_MODE=live

WHATSAPP_NUMBER=+919876543210
```

Appendix: Complete Database Table Index

#	TABLE NAME	PURPOSE
1	<code>users</code>	System user accounts & authentication
2	<code>students</code>	Student profiles & admission records
3	<code>staff</code>	Employee HR records

#	TABLE NAME	PURPOSE
4	school_classes	Class & section configuration
5	subjects	Subject catalogue
6	class_subjects	Class ↔ Subject mapping
7	academic_sessions	Academic year management
8	timetables	Weekly class schedules
9	homeworks	Teacher-assigned homework
10	attendances	Student daily attendance
11	staff_attendances	Staff daily attendance
12	qr_attendance_logs	QR scan attendance log
13	exams	Examination master records
14	exam_schedules	Per-class exam schedule
15	exam_results	Student marks & grades
16	fee_types	Fee category definitions
17	student_fees	Individual fee records & payments
18	fee_discounts	Scholarship & concession records
19	transactions	General income/expense ledger
20	accounts_heads	Chart of accounts
21	salary_records	Monthly payroll records
22	payslip_items	Individual payslip line items
23	library_books	Book catalogue & inventory
24	book_issues	Book issue & return ledger
25	transport_routes	Bus route master
26	transport_stops	Bus stop schedule
27	student_transports	Student ↔ Route assignment
28	hostels	Hostel building records
29	hostel_rooms	Room configuration
30	student_hostels	Student room allotment
31	calendar_events	Annual academic calendar
32	download_materials	Study material uploads
33	live_classes	Live online class schedule
34	notices	School circular notices
35	messages	Internal staff messaging
36	notification_logs	SMS/Email/Push log

#	TABLE NAME	PURPOSE
37	leave_applications	Staff & student leave requests
38	permissions	Permission definitions (44 permissions)
39	role_permissions	Role ↔ Permission mapping
40	custom_fields	Dynamic admission form fields
41	student_custom_field_values	Custom field data per student
42	student_documents	Scanned documents per student
43	student_promotions	Year-end promotion records
44	student_siblings	Sibling relationship for discounts
45	student_timelines	Student academic timeline events
46	student_notes	Teacher notes on students
47	school_settings	System-wide configuration
48	personal_access_tokens	Bearer token store
49	password_reset_tokens	Password reset links
50	sessions	PHP session storage
51	cache	Application cache store
52	cache_locks	Cache lock management
53	jobs	Background job queue
54	job_batches	Batch job tracking
55	failed_jobs	Failed job archive
56	migrations	Database version tracking

Report generated: September 29, 2026 | Infosof Technologies | Smart School Management System v1.5